Market Signals and the Effects of Nominal Disturbances

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Introduction

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Lucas (2003): reaction

> *possibly promising ...[but] to save this particular boat from sinking, we need to throw overboard the islands, rational expectations, utility maximization, cleared competitive markets, and money!*

This paper

Lucas vs. Woodford: Round 2

Question: What happens when we bring back market signals?

Important difference: monopolistic competition
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Answer: Back to Lucas, but with a twist.

Woodford model

Monopolistically competitive supplier $i \in [0,1]$ faces demand curve

$$y_{it} = -\theta(p_{it} - p_t) + y_t, \quad \theta > 1, \tag{1}$$

where

$$p_t = \int_0^1 p_{it} di \quad \text{and} \quad y_t = \int_0^1 y_{it} di.$$

Optimally sets price according to Lerner formula

$$p_{it} = E_{it}[p_t + \zeta y_t], \quad 0 < \zeta \leq 1. \tag{2}$$

Rational expectations given observation vector s_{it}

$$E_{it} = E[\cdot | s_i^t].$$

Equilibrium

Monetary policy maintains exogenous path for nominal output gap

$$q_t = p_t + y_t \tag{3}$$

Definition

An equilibrium is a pair of stationary processes $\{p_{it}, y_{it}\}$ for each supplier satisfying (1), (2), and (3), given stationary $\{q_t\}$, and s_{it} measurable with respect to the history of exogenous disturbances.

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Assumptions: Suppliers' private information u_{it} (an element of s_{it}) and q_t satisfy

$$u_{it} = q_t + \varepsilon_{it}, \qquad q_t = \rho q_{t-1} + v_t \tag{4}$$

with $|\rho| \leq 1$, $\varepsilon_{it} \overset{\text{i.i.d.}}{\sim} N(0, \sigma_\varepsilon^2)$, $v_t \overset{\text{i.i.d.}}{\sim} N(0, \sigma_v^2)$, $\sigma_\varepsilon, \sigma_v > 0$

Lagged full communication

Definition

A lagged full communication equilibrium (lagged FCE) is an equilibrium with

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Proposition 1. The lagged FCE exists and is unique. Moreover,

$$p_t = \gamma q_t + (1-\gamma)\rho q_{t-1}, \qquad y_t = (1-\gamma)v_t,$$

where

$$\gamma \equiv \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\varepsilon^2/\zeta}.$$

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Intuition:

(i) noisier private info or more strategic complementarity implies greater weight on lagged public information.

(ii) purely transitory real effects of nominal disturbances.

Proof of Proposition 1

By (2) and (3),

$$p_t = \bar{E}_t[(1-\zeta)p_t + \zeta q_t]. \tag{5}$$

By recursive substitution, stationarity, and $\zeta \leq 1$,

$$p_t = \sum_{j=1}^{\infty} \zeta(1-\zeta)^{j-1} \bar{E}_t^{(j)} q_t.$$

By (4),

$$\bar{E}^{(j)} q_t = g^j q_t + (1-g^j)\rho q_{t-1},$$

where $g = \sigma_v^2/(\sigma_v^2 + \sigma_\varepsilon^2)$ is the gain parameter.

Substitute into the previous expression for p_t and evaluate series to obtain the law of motion stated in the proposition. $\square$

Lagged endogenous signals

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Proposition 2. The lagged FCE is a REE if and only if

$$\frac{1}{\theta - 1}\left(1 + \frac{\sigma_v^2}{\sigma_\varepsilon^2/\zeta}\right) + \frac{\sigma_v^2}{\sigma_\varepsilon^2/\zeta} \geq |\rho|.$$

Proof of Proposition 2

Proof. Supplier output is:

$$y_{it} = -\theta(p_{it} - p_t) + (q_t - p_t).$$

Since p_{it} is known to firm, y_{it} has same info $n_t \equiv (\theta - 1)p_t + q_t$.

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For FCE to be REE, <u>n_t must be invertible.</u> In the lagged FCE,

$$n_t = [1 + (\theta - 1)\gamma + (\theta - 1)(1 - \gamma)\rho L]q_t \equiv A(L)q_t.$$

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Expression comes from rearranging above condition. $\square$

Intuition

Recall condition:

$$\frac{1}{\theta - 1}\left(1 + \frac{\sigma_v^2}{\sigma_\varepsilon^2/\zeta}\right) + \frac{\sigma_v^2}{\sigma_\varepsilon^2/\zeta} \geq |\rho|.$$

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Recall condition:

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Without enough weight on private info, n_{t-1} doesn't reveal enough about last period's disturbance.

Cannot support FCE.

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Cannot support FCE.

Proposition 3.1 of Huo and Pedroni (2023, JME) when $\theta \to \infty$.
$\hookrightarrow$ observe lagged p rather than y_i.

Uniqueness and existence

Proposition 3. If $s_{it} = (u_{it}, y_{i,t-1})$, the lagged FCE is the only possible linear symmetric REE.

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Proof.

1. Show in any linear symmetric REE, $n_t = A(L)q_t$.

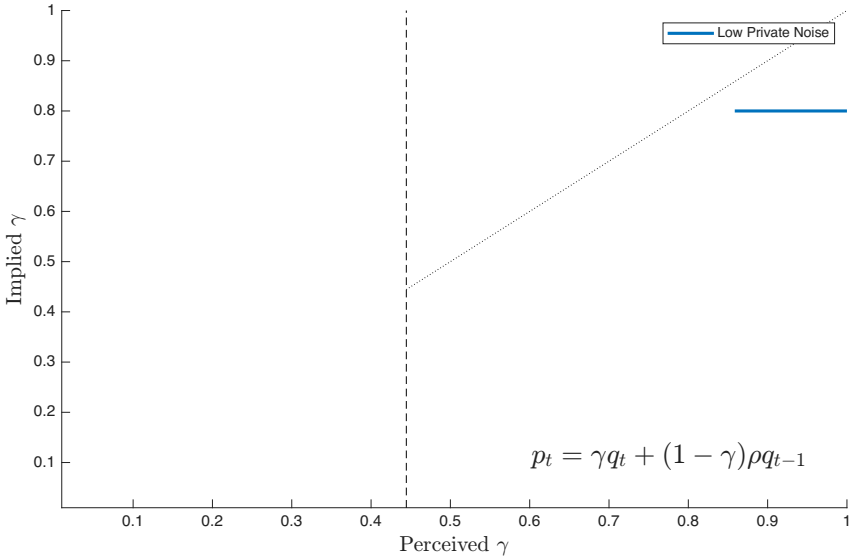
2. Find Wold representation $s_{it} = \Gamma(L)w_{it}$ in closed form.

3. Use $\Gamma(L)$ and Wiener-Kolmogorov filter to obtain equilibrium fixed-point equation $A(L) = T[A(L)]$ in closed form.

4. Show $A(L)$ must be invertible to satisfy this equation. □

Corollary. If the lagged FCE is not a REE, then no linear symmetric REE exists. (The Twist!)

Some Graphical Intuition - Part I



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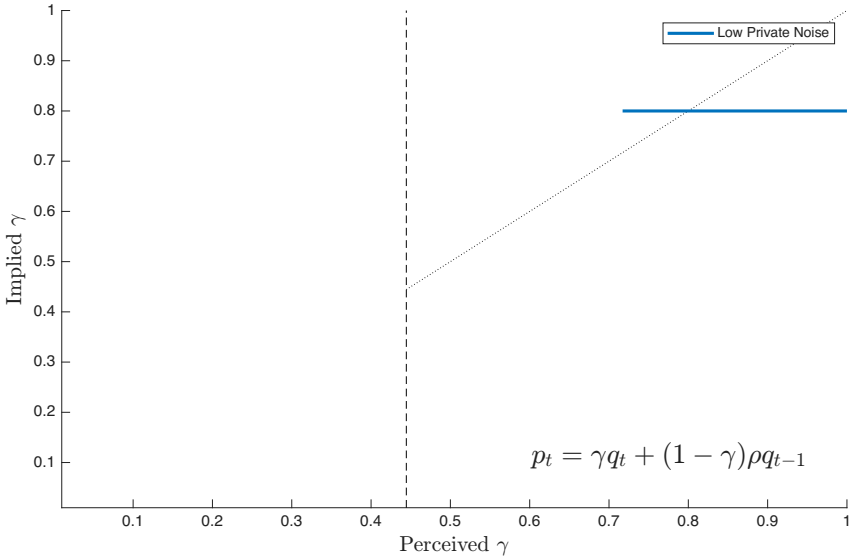

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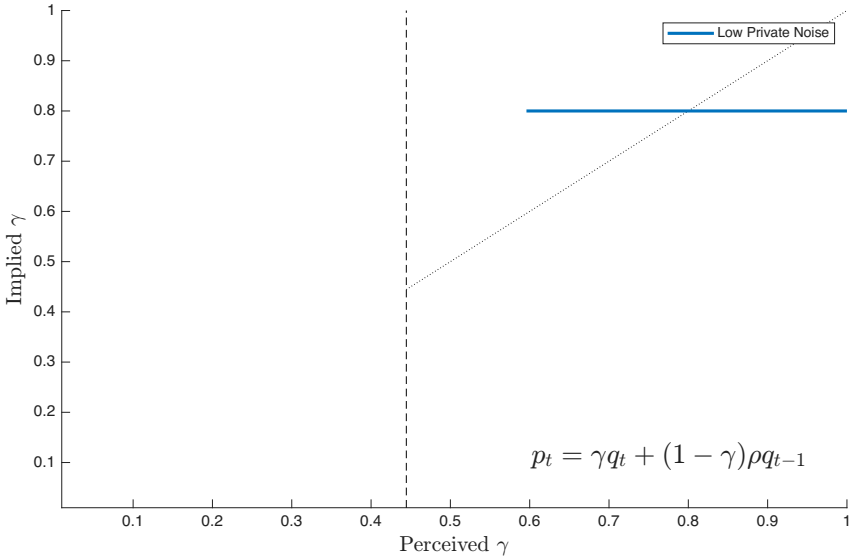

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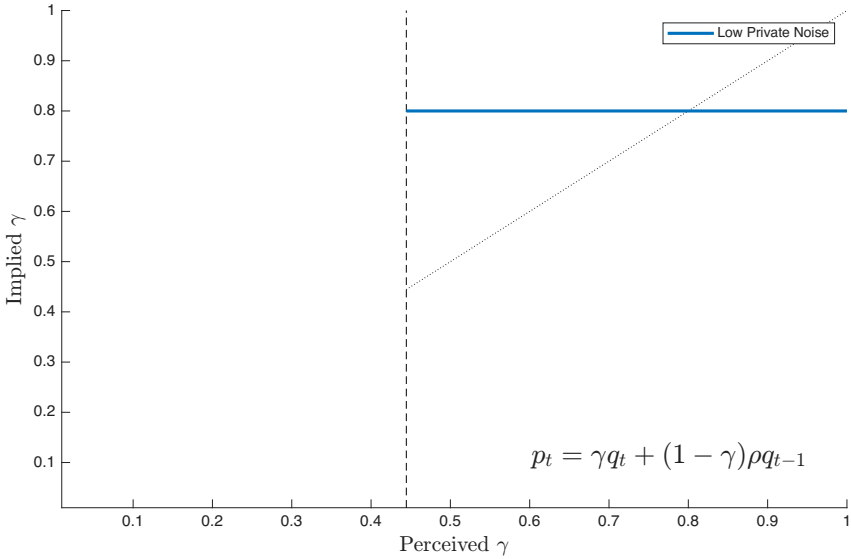

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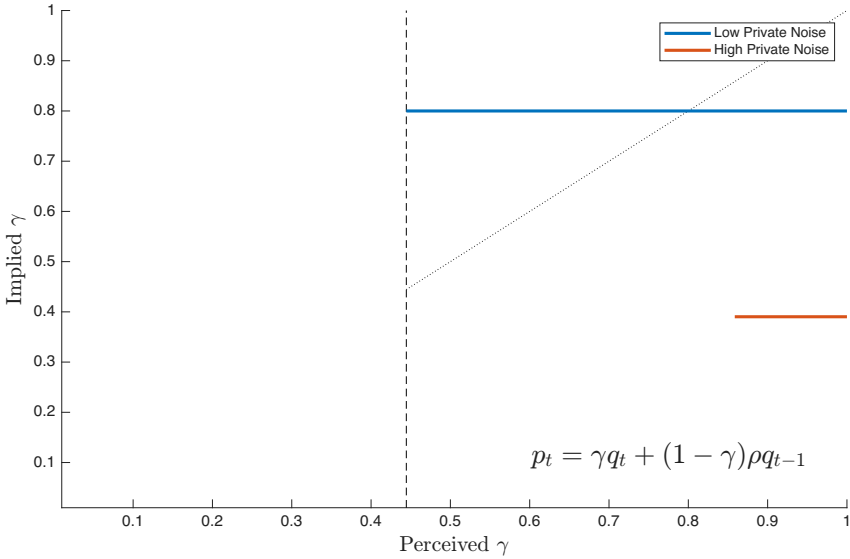

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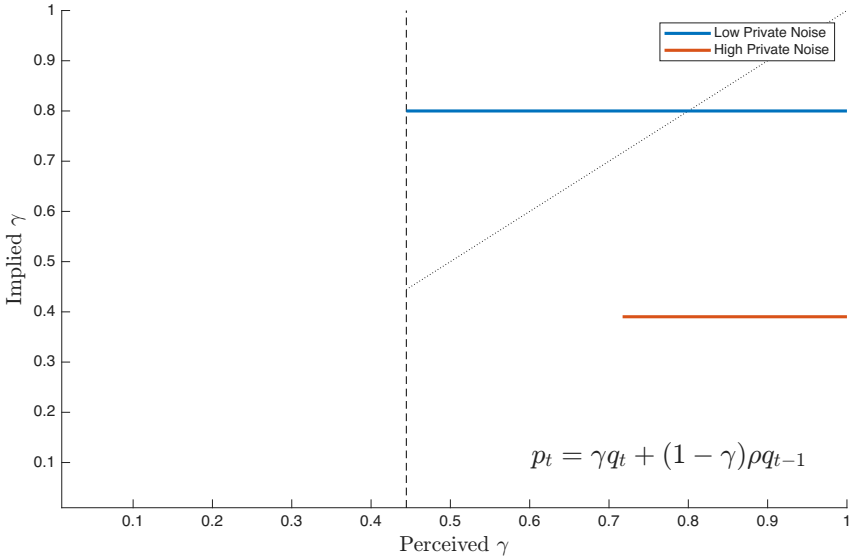

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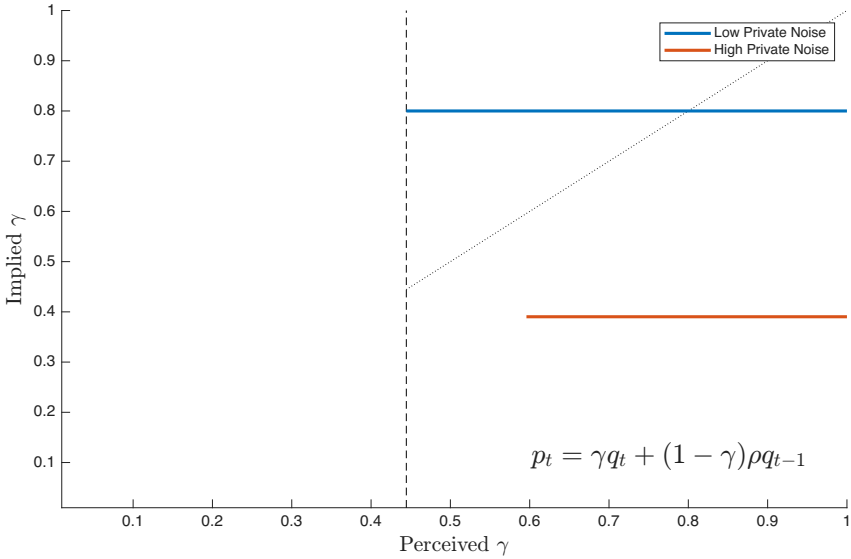

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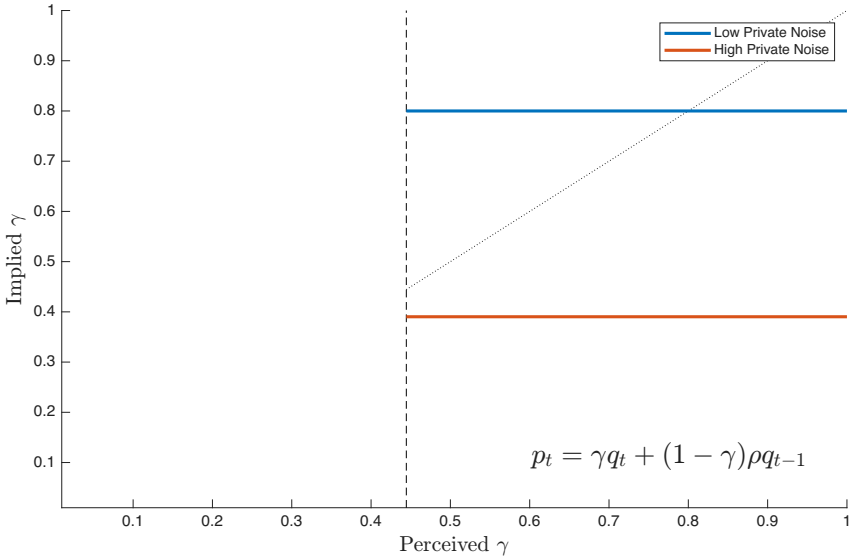

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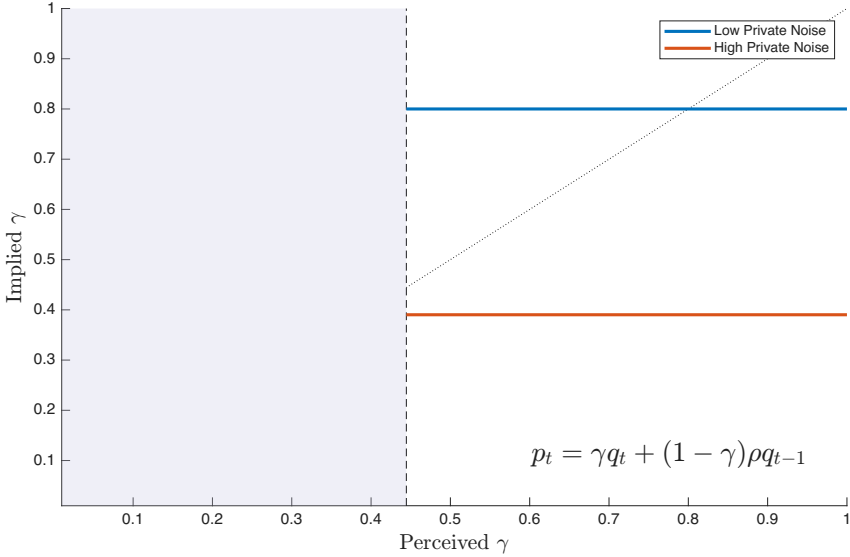

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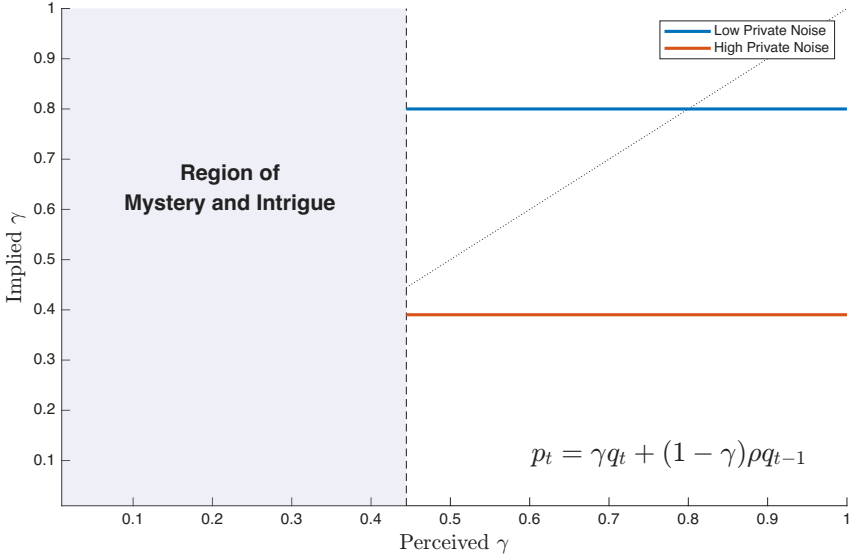

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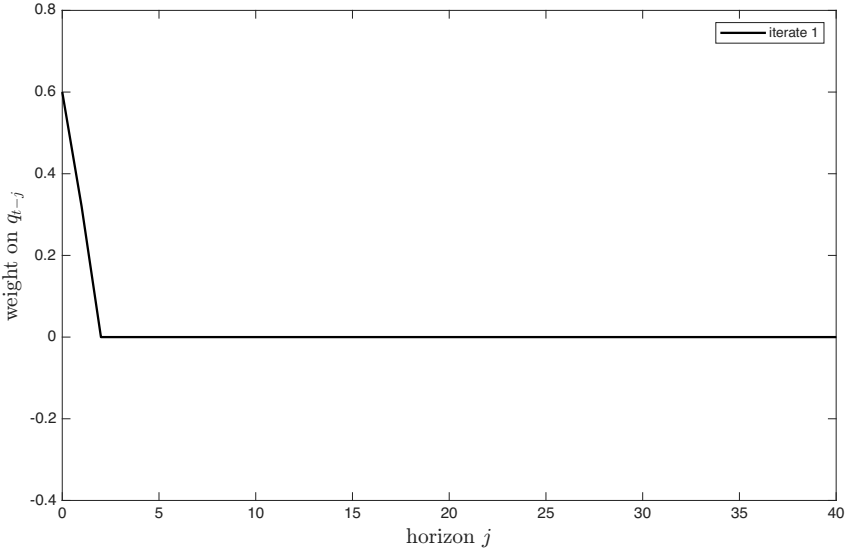

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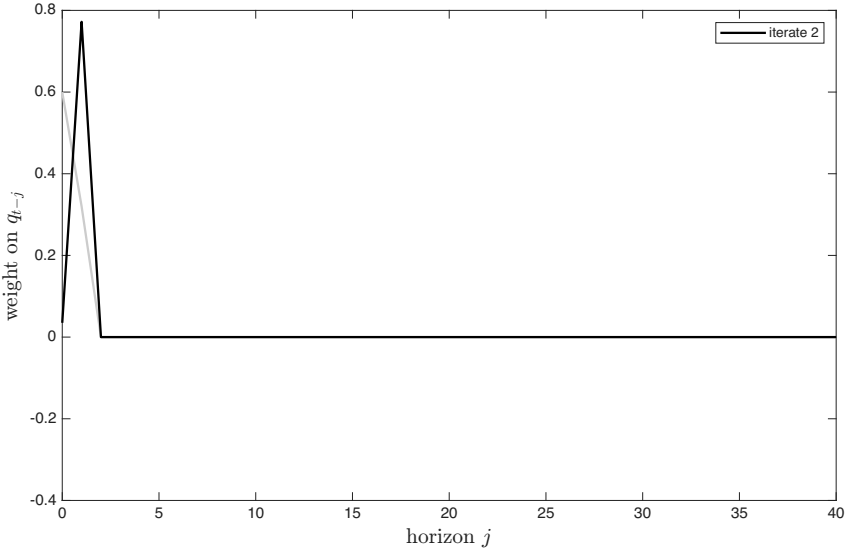

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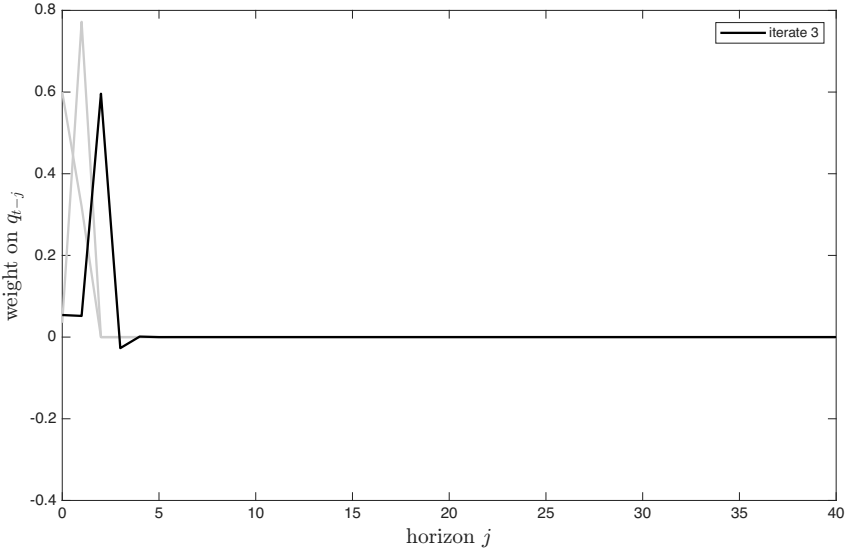

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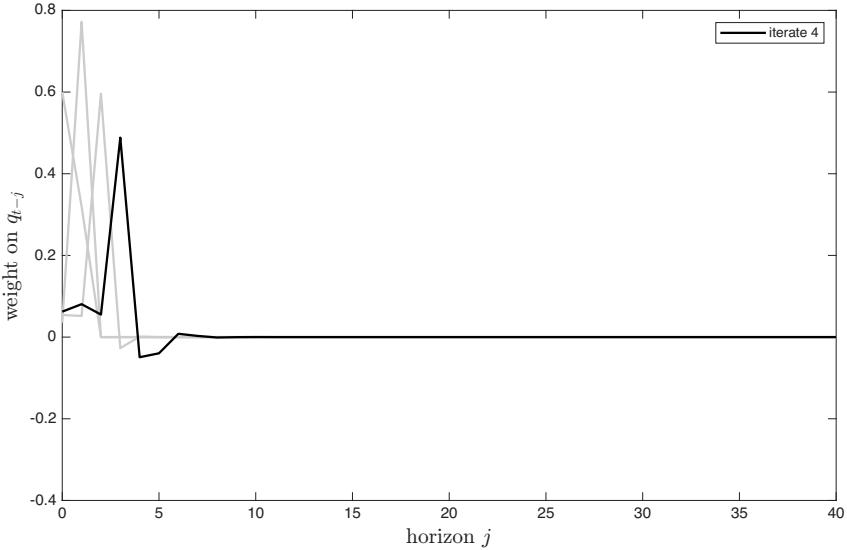

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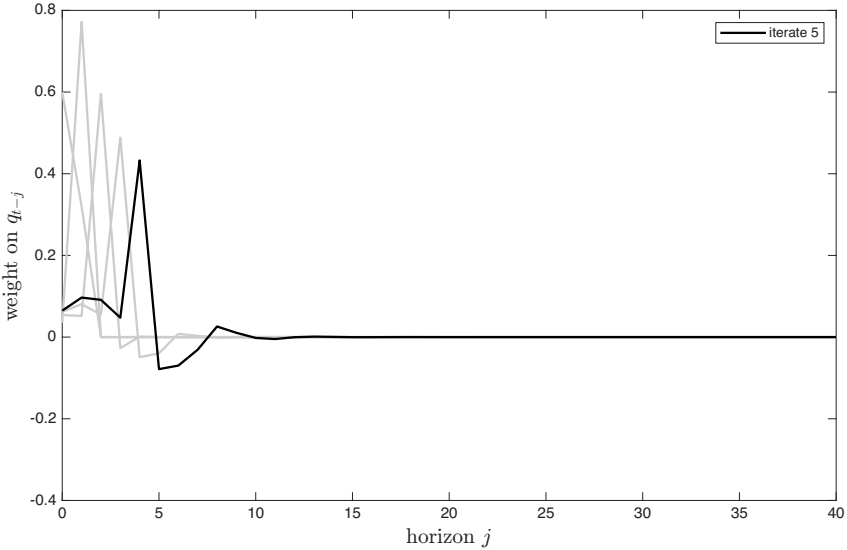

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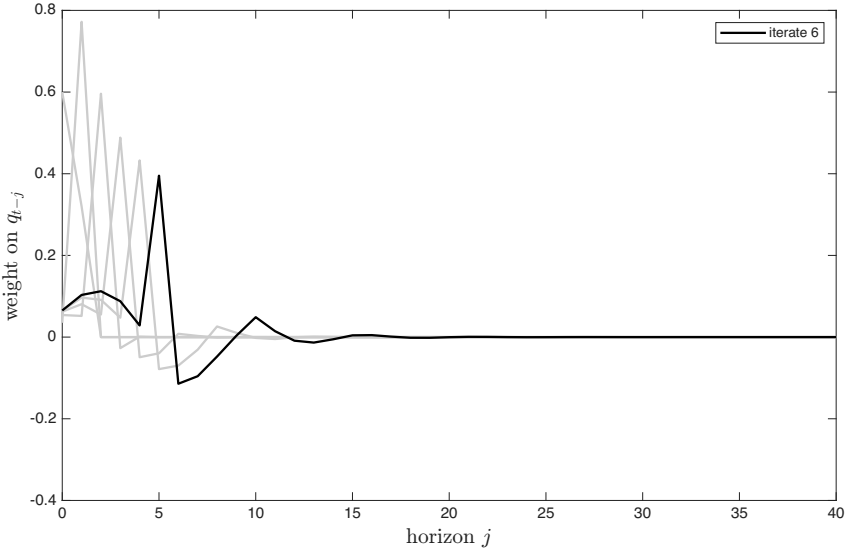

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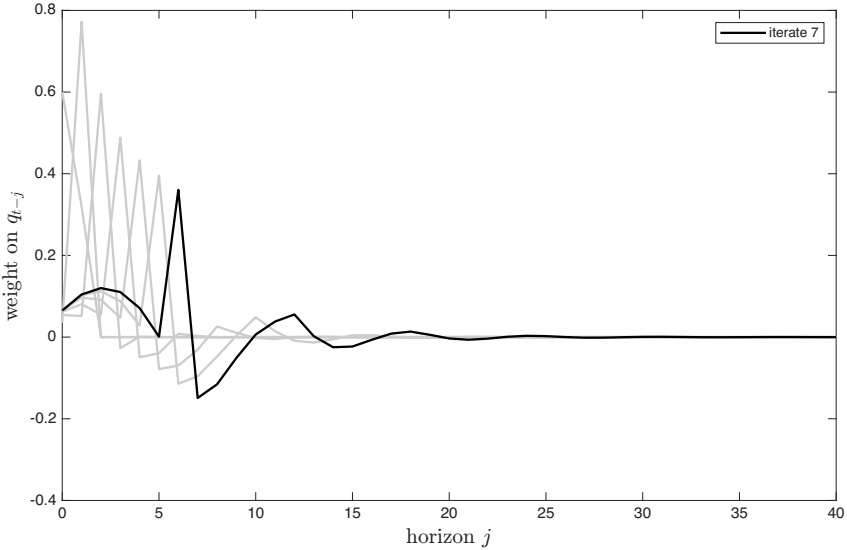

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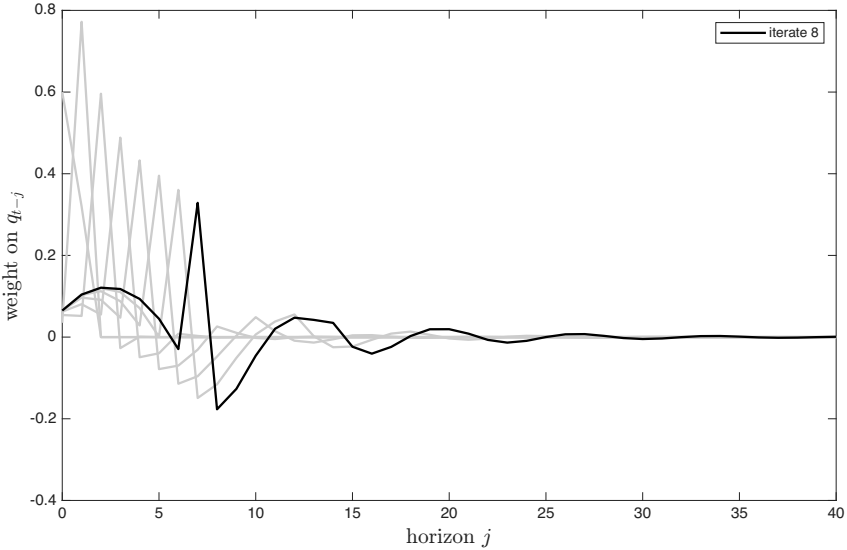

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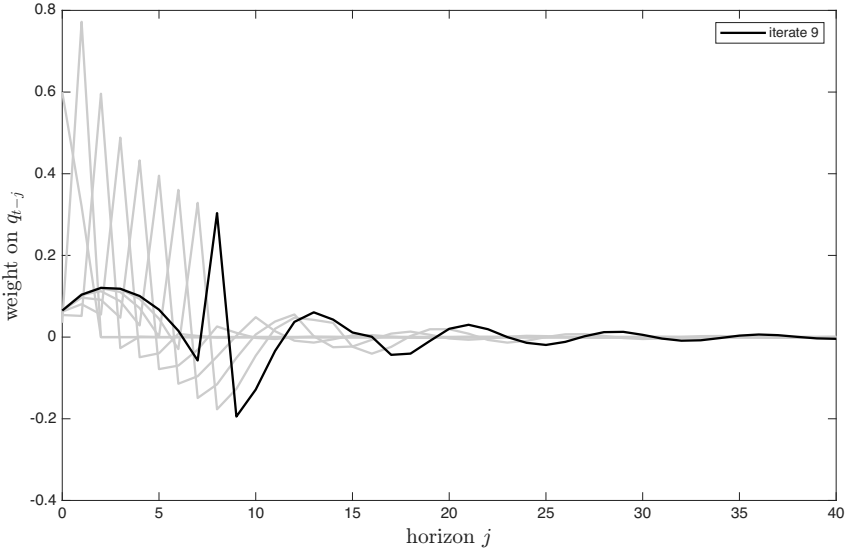

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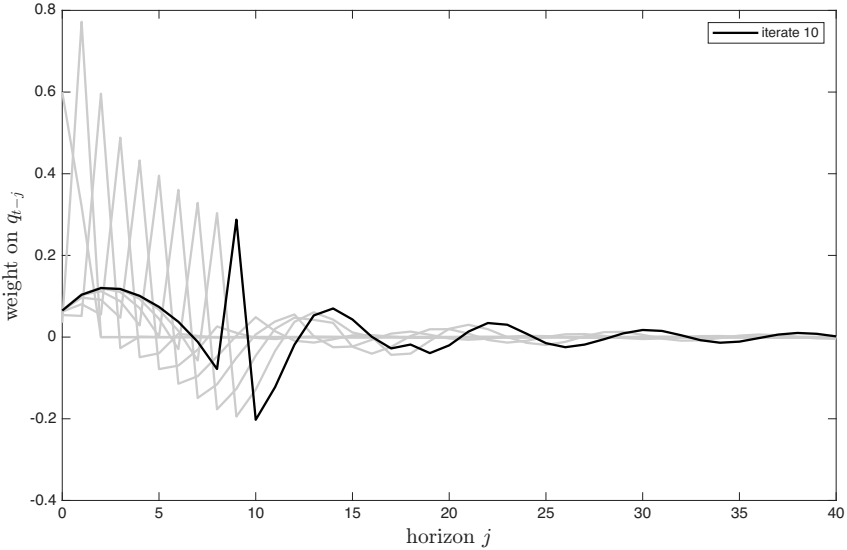

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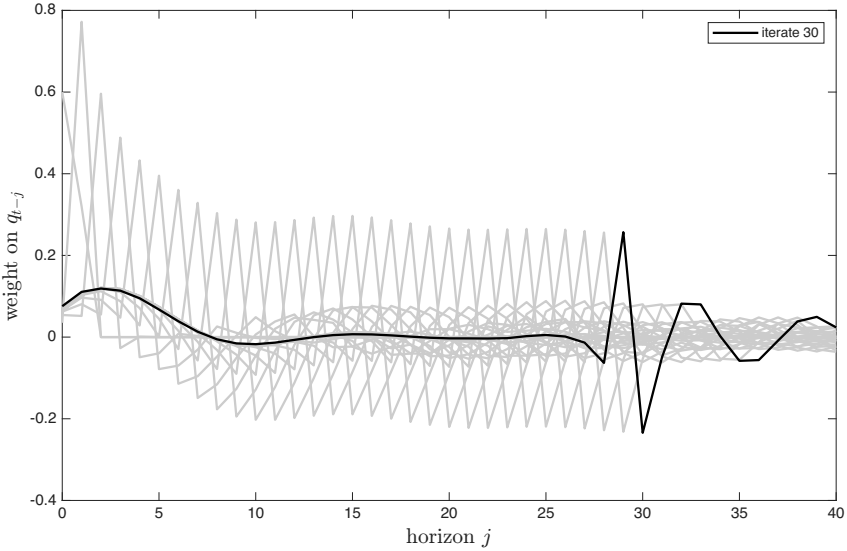

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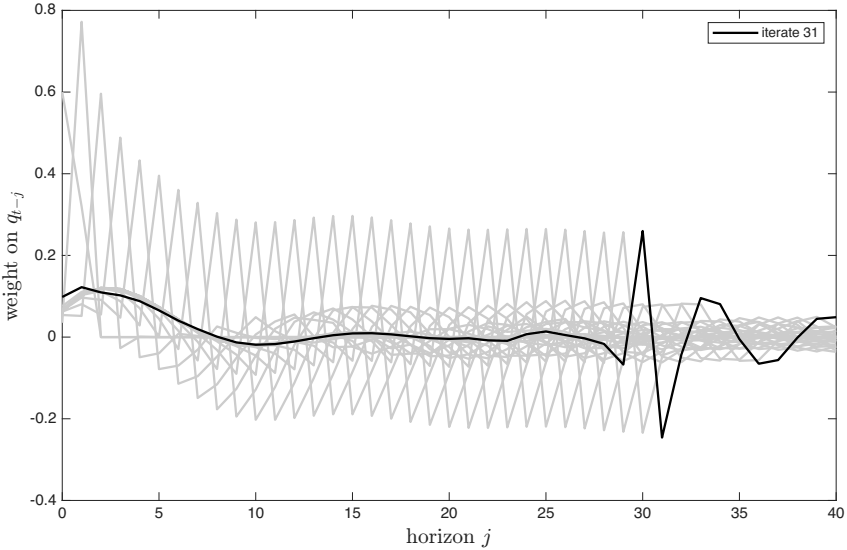

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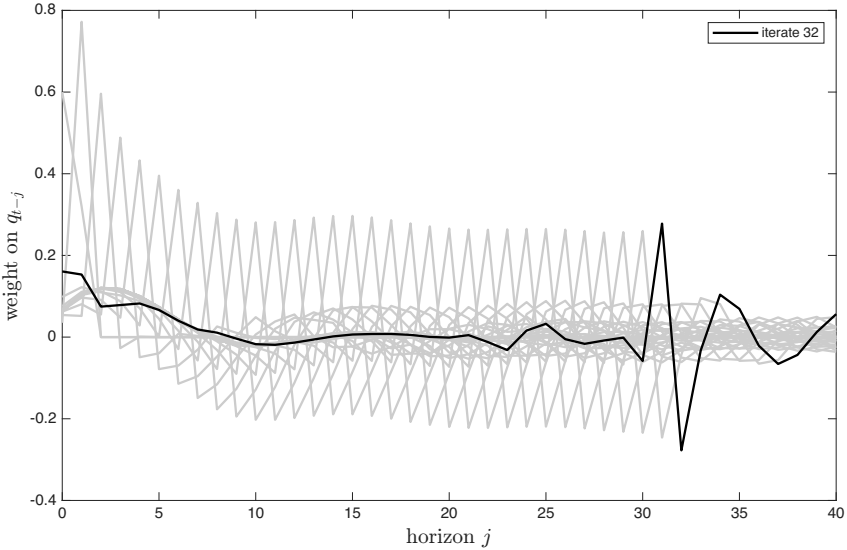

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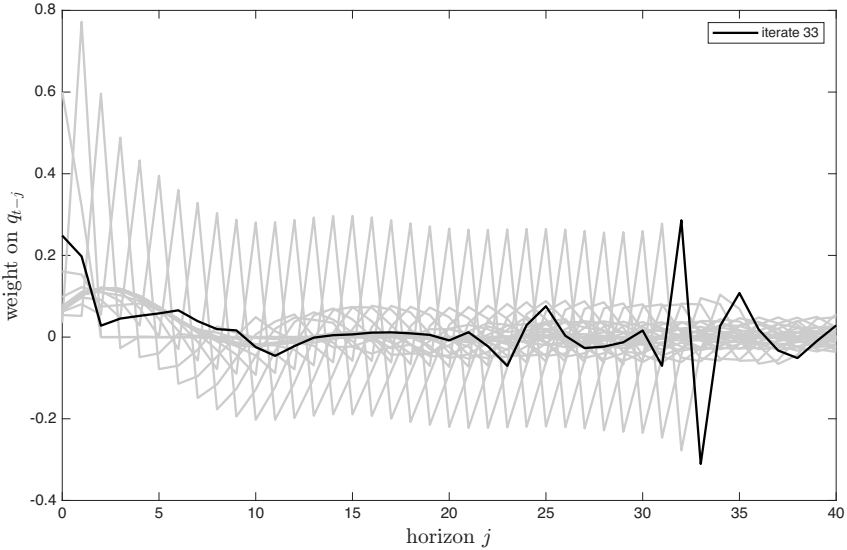

Literature

Two-period futures market example in Section 4 of Kreps (1977) on "fulfilled expectations" equilibria.

Dynamic asset pricing example in Section 5 of Futia (1981): "This connection between the dynamic stability of RE price processes and the amount of information they reveal seems to persist in other linear models and may be a quite general phenomenon" (p.186).

Rondina and Walker (2013) argue this phenomenon is impossible: (a) if a REE exists, then disturbances are known, (c) therefore, disturbances are known and there exists a unique REE (the FCE).

Huo and Pedroni (2023) refer to the non-invertible regime as the "confounding regime," where nominal disturbances (a) have persistent effects, unlike in the FCE, and (b) produce initial under-reaction and delayed over-reaction relative to the FCE.

General lag length

Lagged full communication

Definition

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A k-lagged FCE is a REE with $s_{it} = (u_{it}, q_{t-k})$.

Proposition 4. The k-lagged FCE exists and is unique. In this equilibrium

$$p_t = \bar{B}(L)q_t \quad \text{and} \quad y_t = (1 - \bar{B}(L))q_t$$

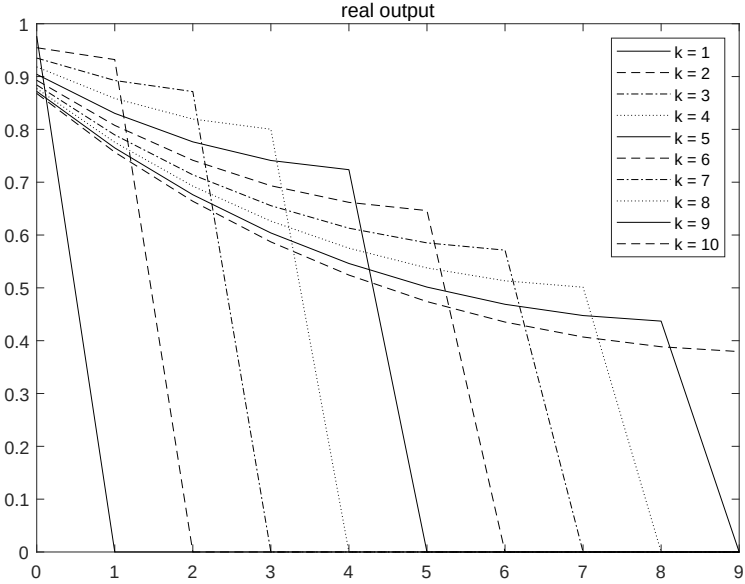
where

$$\bar{B}(L) = \frac{(1-r^2)r^{k+1}(1-\rho L)(L-\rho)L^k + (1-r\rho)(\rho-r)[r^{2k+1}(1-rL)+(L-r)]}{\rho[(1-r\rho)+(\rho-r)r^{2k+1}](1-rL)(L-r)}$$

and $0 < |r| < |\rho| < 1$ solves

$$\rho r^2 - \left(1 + \rho^2 + \zeta\frac{\sigma_v^2}{\sigma_\varepsilon^2}\right) r + \rho = 0.$$

Numerical example: $(\rho, \sigma_v^2/\sigma_\varepsilon^2) = (1, 0.16)$



Lagged endogenous aggregate signal

Proposition 5. Consider an economy in which $s_{it} = (u_{it}, y_{i,t-k})$. In any linear symmetric REE,

(a) the paths of $\{p_{it}, y_{it}\}$ are the same as in the k-lagged FCE, and

(b) this equilibrium exists if and only if $\bar{A}(L) = 1 + (\theta - 1)\bar{B}(L)$ is invertible.

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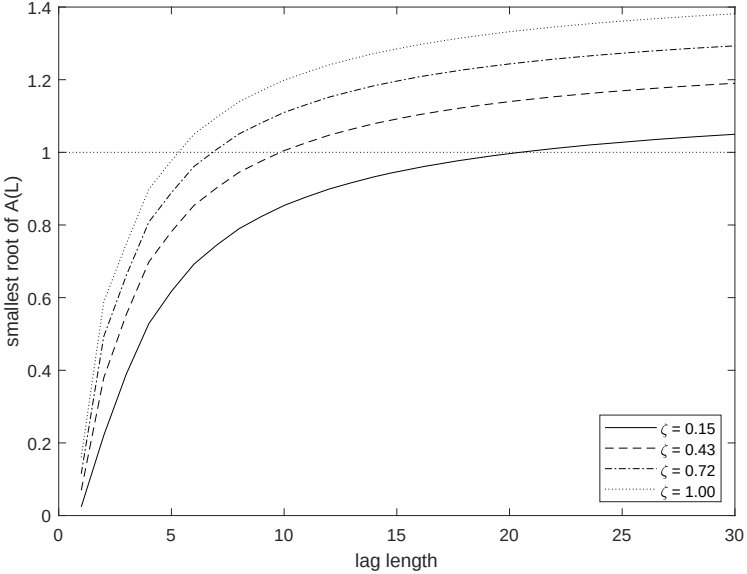
Intuition: a longer lag makes knowledge of lagged disturbances less useful, so suppliers put more weight on their private info, making the endogenous aggregate more informative.

Related to literature on welfare losses from releasing public information; e.g., Amador and Weill (2010).

Numerical example: $(\rho, \sigma_v^2/\sigma_\varepsilon^2) = (1, 0.16)$



Imperfect perception

Imperfect perception of endogenous aggregates

Distinction: availability vs awareness of public information.

So far, we have not maintained this distinction with respect to endogenous aggregate information.

To do so, suppose now that

$$s_{it} = (u_{it}, \tilde{y}_{i,t-k}),$$

where the endogenous signal is imperfectly perceived:

$$\tilde{y}_{it} = y_{it} + \eta_{it}, \quad \eta_{it} \overset{\text{i.i.d.}}{\sim} N(0, \sigma_\eta^2).$$

Results with imperfect perception

Proposition 6. Consider an economy in which $s_{it} = (u_{it}, \tilde{y}_{i,t-k})$. In any linear symmetric REE,

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Results with imperfect perception

Proposition 6. Consider an economy in which $s_{it} = (u_{it}, \tilde{y}_{i,t-k})$. In any linear symmetric REE,

(a) $n_t = A(L)q_t$, with $A(L)$ invertible, and

(b) $A(L)$ is not finite-order ARMA.

Results with imperfect perception

Proposition 6. Consider an economy in which $s_{it} = (u_{it}, \tilde{y}_{i,t-k})$. In any linear symmetric REE,

(a) $n_t = A(L)q_t$, with $A(L)$ invertible, and

(b) $A(L)$ is not finite-order ARMA.

(c) If $\bar{A}(L)$ is invertible, there exists a unique solution for $A(L)$ for all σ_η in a neighborhood of $\sigma_\eta = 0$, and

$$A(L) = \bar{A}(L) + \mathcal{O}(\sigma_\eta^2).$$

Proof. Exactly as in Proposition 3, find fixed point equation:

$$A(L) = \frac{[\phi^2(1-\rho r)(1-rL) + (\rho - r)(L-r)]H(L)}{\rho(1+\phi^2)(1-rL)(L-r)H(L) - \phi\frac{\sigma_\varepsilon}{\sigma_v}(1-r^2)(1-\rho L)(L-\rho)L},$$

$$\phi = \sqrt{\frac{\rho - r}{1-\rho r}}\, r^{k+1/2}\frac{A(r)}{H(r)}, \quad |H(L)|^2 = |A(L)|^2 + \frac{\sigma_\eta^2}{\sigma_v^2}\frac{\rho}{r}|1-rL|^2.$$

(a) Use Rouche's theorem to show that $A(L)$ has no inside zeros.

(b) Suppose $A(L) = \frac{P(L)}{Q(L)}$, derive contradiction.

(c) Rewrite as $F(A, \sigma_\eta) = 0$ and use implicit function theorem. If $F(\bar{A}, 0) = 0$, then

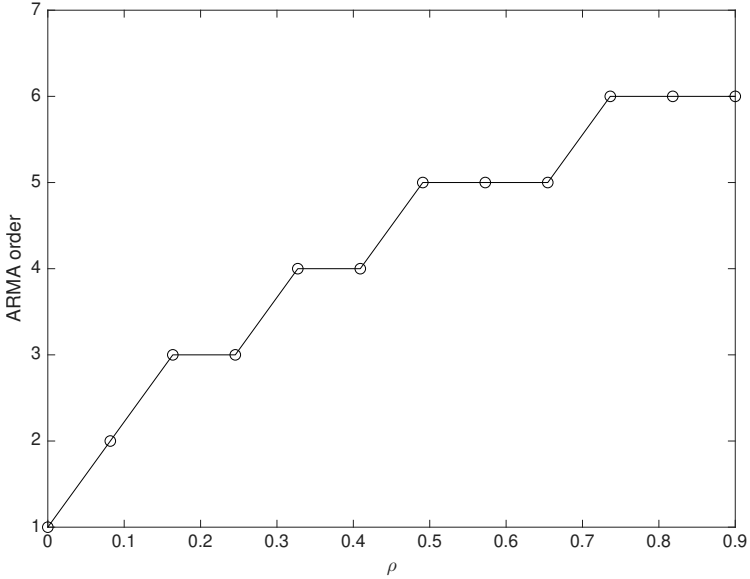
$$F_A(\bar{A}, 0) = 1 \quad \text{and} \quad F_{\sigma_\eta}(0, \bar{A}) = 0,$$

so F_A is invertible and $A = \bar{A} - F_A^{-1}F_{\sigma_\eta}\sigma_\eta + \mathcal{O}(\sigma_\eta^2)$. $\square$

Literature

First example (that we know of) that proves existence of a infinite order equilibrium.

ARMA approximations: $(\zeta, \sigma_v^2/\sigma_\varepsilon^2, \sigma_\eta^2) = (1, 1, 0.05)$

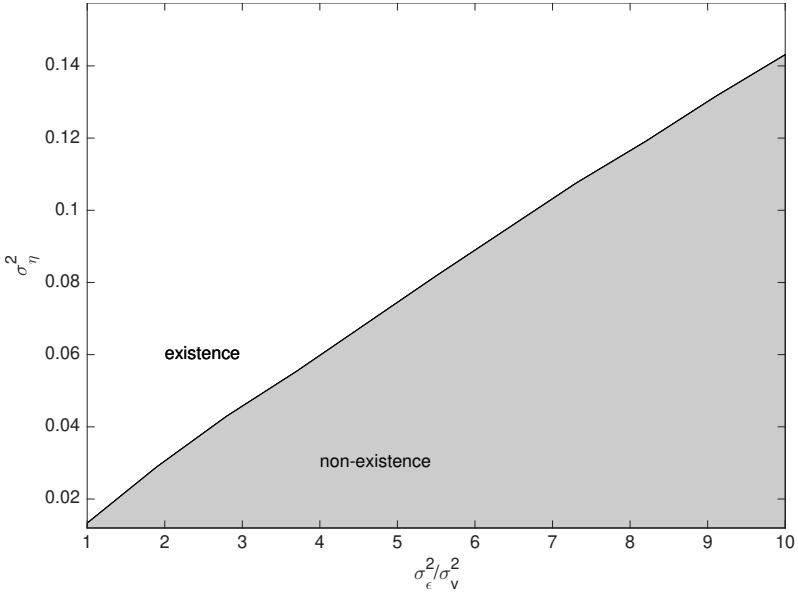


Informed conjecture

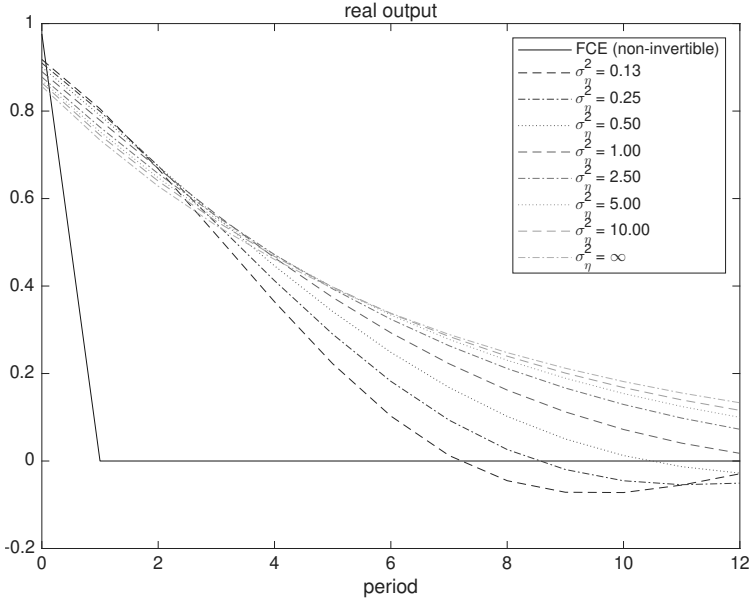
When $\bar{A}(L)$ is not invertible, there exists a finite level of perceptual noise (σ_η) such that an equilibrium does exist.

1. Intuition: non-existence is about too much weight on private signal.
2. Fact: when $\sigma_\eta \to \infty$, we know there is an equilibrium.

If $\bar{A}(L)$ is not invertible: $(\rho, \zeta) = (1, 0.15)$



Effect of nominal disturbance: $(\rho, \sigma_v^2/\sigma_\varepsilon^2) = (1, 0.16)$



Summary and work in progress

1. Endogenous aggregates fully reveal nominal disturbances if suppliers respond strongly enough to their private information.

2. If not, the equilibrium can fail to exist. Important to realize this!

3. Imperfect perception of endogenous aggregates makes existence "more likely," and precludes full revelation after a lag.

4. We provide the first proof of the (local) existence and uniqueness of an equilibrium with non-ARMA dynamics.

In progress:

a. Restore local-aggregate misperception as in Lucas (1972).

b. Empirical application to micro data.

References

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Appendix

Private information benchmark

Proposition. Consider an economy in which $s_{it} = u_{it}$. There exists a unique REE, in which

$$p_t = \frac{(1 - r/\rho)}{(1 - rL)} q_t, \qquad y_t - y_t^n = \frac{r}{\rho} \frac{(1 - \rho L)}{(1 - rL)} q_t$$

where $0 < r < \rho$ solves

$$\rho r^2 - \left(1 + \rho^2 + \frac{\sigma_v^2}{\sigma_\varepsilon^2} \zeta\right) r + \rho = 0.$$

Step 1: show that $p_t = A(L)q_t$

(a) Since p_{it} is measurable with respect to the history of $s_{it} = (u_{it}, p_{t-1})$, in any linear symmetric REE

$$p_{it} = B_u(L)u_{it} + B_p(L)Lp_t,$$

where $B_u(L)$ and $B_p(L)$ are one-sided into the past.

(b) Average across i to get

$$p_t = B_u(L)q_t + B_p(L)Lp_t.$$

(c) Solve for p_t

$$p_t = \frac{B_u(L)}{1 - B_p(L)L} q_t \equiv A(L)q_t,$$

where $A(L)$ is causal by definition of REE.

Step 2: find Wold representation $s_{it} = \Gamma(L) w_{it}$

Write spectral density of s_{it} in terms of $A(L)$ and perform spectral factorization $f(\lambda) = \frac{1}{2\pi}\Gamma(e^{-i\lambda})\Gamma(e^{-i\lambda})^*$ following Rozanov (1967).

$$\Gamma(L) = \frac{1}{\sqrt{1+\phi^2}} \frac{\sigma_v}{(1-\rho L)} \sqrt{\frac{r}{\rho}} \times$$

$$\begin{bmatrix} \phi \frac{\rho}{r}(L-r) & -\frac{\rho}{r}\frac{\sigma_\varepsilon}{\sigma_v}(1-rL) \\ \frac{\phi \frac{\sigma_v}{\sigma_\varepsilon} A(L) L^2 + (1-\rho L) H(L)}{(1-rL)} & -\frac{\frac{\sigma_v}{\sigma_\varepsilon} A(L) L^2 + \phi(1-\rho L) H(L)}{(L-r)} \end{bmatrix},$$

where

$$\phi \equiv \sqrt{\frac{\sigma_v}{\sigma_\varepsilon}} \frac{r^2}{1-\rho r} \frac{A(r)}{H(r)},$$

and $H(L)$ is the Wold factor from $H(L)H(L^{-1}) = A(L)A(L^{-1})$.

Step 3: Find fixed point equation $A(L) = T[A(L)]$.

(a) According to the structural model,

$$p_t = \bar{E}_t[(1-\zeta)p_t + \zeta q_t]$$

(b) By the Wiener-Kolmogorov filter

$$p_t = \left\{ \begin{bmatrix} \frac{((1-\zeta)A(L)+\zeta)\sigma_v}{1-\rho L} & 0 \end{bmatrix} M(L^{-1})'\Gamma(L^{-1})'^{-1} \right\}_+ \Gamma(L)^{-1} M(L) \begin{bmatrix} v_t \\ 0 \end{bmatrix}$$

(c) These imply the fixed-point equation

$$A(L) = \frac{[\phi^2(1-\rho r)(1-rL) + (\rho - r)(L-r)]H(L)}{\rho(1+\phi^2)(1-rL)(L-r)H(L) - \phi\frac{\sigma_\varepsilon}{\sigma_v}(1-r^2)(1-\rho L)(L-\rho)L}.$$

Step 4: show $A(L)$ must be invertible

From steps 1-3, the fixed-point equation is

$$A(L) = \frac{[\phi^2(1-\rho r)(1-rL) + (\rho - r)(L-r)]H(L)}{\rho(1+\phi^2)(1-rL)(L-r)H(L) - \phi\frac{\sigma_\varepsilon}{\sigma_v}(1-r^2)(1-\rho L)(L-\rho)L},$$

$$\phi = \sqrt{\frac{\rho - r}{1-\rho r}}\, r^{k+1/2}\,\frac{A(r)}{H(r)}, \quad H(L)H(L^{-1}) = A(L)A(L^{-1}).$$

(a) Show by substitution that $A(r) \neq 0$.

(b) Suppose $A(\alpha) = 0$ for inside zero $\alpha \neq r$ (at most one).

(b) Implies $\phi^2(1-\rho r)(1-r\alpha) + (\rho - \alpha)(\alpha - r) = 0$.

(c) By $\frac{A(r)}{H(r)} = \frac{r-\alpha}{1-\alpha r}$ and definition of ϕ, implies the contradiction

$$(\rho - r)(\alpha - r)(1 - r^{2k+1}) = 0.$$